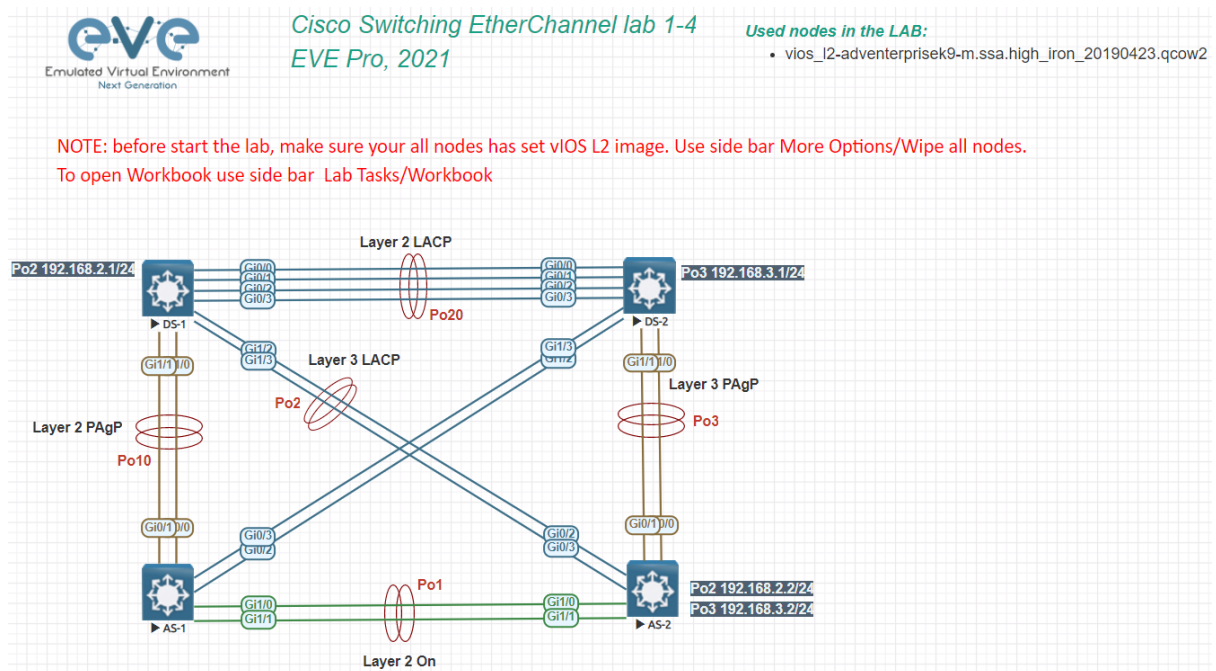


## CCNP Switching EtherChannels Lab

### Topology



### Objectives

1. Configure Basic Device Settings
2. Configure Layer 2 EtherChannel mode On
3. Configure Layer 2 EtherChannel PAGP
4. Configure Layer 2 EtherChannel LACP
5. Configure Layer 3 EtherChannel PAGP
6. Configure Layer 3 EtherChannel LACP
7. Create static routes

An EtherChannel can be established using one of three mechanisms:

- **PAgP** - Cisco's proprietary negotiation protocol;
- **LACP (IEEE 802.3ad)** - Standards-based negotiation protocol;
- **Static Persistence ("On")** - No negotiation protocol is used;

| LACP    |        |         | PAgP      |           |      |
|---------|--------|---------|-----------|-----------|------|
|         | Active | Passive |           | Desirable | Auto |
| Active  | Yes    | Yes     | Desirable | Yes       | Yes  |
| Passive | Yes    | No      | Auto      | Yes       | No   |

### Task 1: Configure Basic

#### Switch DS-1

```
hostname DS-1
spanning-tree mode rapid-pvst
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g2/0-3
shutdown
exit
```

#### Switch DS-2

```
hostname DS-2
spanning-tree mode rapid-pvst
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g1/2-3, g2/0-3
shutdown
exit
```

#### Switch AS-1

```
hostname AS-1
spanning-tree mode rapid-pvst
line con 0
exec-timeout 0 0
logging synchronous
exit
interface range g0/2-3, g1/2-3
shutdown
exit
```

#### Switch AS-2

## CCNP Switching EtherChannels Lab

```
hostname AS-2
spanning-tree mode rapid-pvst
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
interface range g1/2-3
  shutdown
  exit
```

### Task 2: Configure Layer 2 EtherChannel mode On

#### Switch AS-1

```
interface range g1/0-1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  shutdown
  channel-group 1 mode on
  no shutdown
```

#### Switch AS-2

```
interface range g1/0-1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  shutdown
  channel-group 1 mode on
  no shutdown
```

#### Verification:

**AS-1#show etherchannel summary**

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:           1
```

## CCNP Switching EtherChannels Lab

| Group              | Port-channel | Protocol | Ports     |           |
|--------------------|--------------|----------|-----------|-----------|
| -----+-----+-----+ |              |          |           |           |
| 1                  | Po1 (SU)     | -        | Gi1/0 (P) | Gi1/1 (P) |

### Task 3: Configure Layer 2 EtherChannel PAGP

#### Switch DS-1

```
interface range g1/0-1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  shutdown
  channel-protocol pagp
  channel-group 10 mode desirable
  no shutdown
```

#### Switch AS-1

```
interface range g0/0-1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  shutdown
  channel-protocol pagp
  channel-group 10 mode auto
  no shutdown
```

### Verification:

```
DS-1#show etherchannel 10 port
      Ports in the group:
      -----
Port: Gi1/0
-----
Port state      = Up Mstr In-Bndl
Channel group = 10          Mode = Desirable-S1      Gcchange = 0
Port-channel   = Po10      GC   = 0x000A0001        Pseudo port-channel = Po10
Port index     = 0         Load = 0x00              Protocol =  PAgP

Flags:  S - Device is sending Slow hello.  C - Device is in Consistent state.
        A - Device is in Auto mode.        P - Device learns on physical port.
        d - PAgP is down.

Timers: H - Hello timer is running.        Q - Quit timer is running.
        S - Switching timer is running.    I - Interface timer is running.

Local information:

      Hello   Partner  PAgP    Learning  Group
Port   Flags State  Timers  Interval Count  Priority  Method  Ifindex
Gi1/0   SC   U6/S7   H       30s      1       128     Any     14
```

## CCNP Switching EtherChannels Lab

Partner's information:

|       | Partner | Partner        | Partner | Partner Group  |
|-------|---------|----------------|---------|----------------|
| Port  | Name    | Device ID      | Port    | Age Flags Cap. |
| Gi1/0 | AS-1    | 500a.0003.8000 | Gi0/0   | 22s SAC A0001  |

Age of the port in the current state: 0d:00h:05m:27s

Port: Gi1/1

-----

Port state = Up Mstr In-Bndl

Channel group = 10 Mode = Desirable-S1 Gcchange = 0

Port-channel = Po10 GC = 0x000A0001 Pseudo port-channel = Po10

Port index = 0 Load = 0x00 Protocol = PAgP

Flags: S - Device is sending Slow hello. C - Device is in Consistent state.  
A - Device is in Auto mode. P - Device learns on physical port.  
d - PAgP is down.

Timers: H - Hello timer is running. Q - Quit timer is running.  
S - Switching timer is running. I - Interface timer is running.

Local information:

|       |       |       |        | Hello    | Partner | PAgP     | Learning | Group   |
|-------|-------|-------|--------|----------|---------|----------|----------|---------|
| Port  | Flags | State | Timers | Interval | Count   | Priority | Method   | Ifindex |
| Gi1/1 | SC    | U6/S7 | H      | 30s      | 1       | 128      | Any      | 14      |

Partner's information:

|       | Partner | Partner        | Partner | Partner Group  |
|-------|---------|----------------|---------|----------------|
| Port  | Name    | Device ID      | Port    | Age Flags Cap. |
| Gi1/1 | AS-1    | 500a.0003.8000 | Gi0/1   | 19s SAC A0001  |

Age of the port in the current state: 0d:00h:05m:21s

DS-1#show etherchannel 10 detail

Group state = L2

Ports: 2 Maxports = 4

Port-channels: 1 Max Port-channels = 1

Protocol: PAgP

Minimum Links: 0

Ports in the group:

-----

Port: Gi1/0

-----

## CCNP Switching EtherChannels Lab

```
Port state      = Up Mstr In-Bndl
Channel group = 10          Mode = Desirable-S1      Gcchange = 0
Port-channel    = Po10      GC   = 0x000A0001      Pseudo port-channel = Po10
Port index      = 0         Load = 0x00           Protocol = PAgP

Flags: S - Device is sending Slow hello.  C - Device is in Consistent state.
      A - Device is in Auto mode.          P - Device learns on physical port.
      d - PAgP is down.

Timers: H - Hello timer is running.        Q - Quit timer is running.
      S - Switching timer is running.      I - Interface timer is running.

Local information:

      Hello    Partner  PAgP    Learning  Group
Port   Flags State  Timers  Interval Count  Priority  Method  Ifindex
Gi1/0   SC   U6/S7   H       30s      1       128     Any     14

Partner's information:

      Partner          Partner          Partner          Partner Group
Port   Name            Device ID        Port    Age  Flags  Cap.
Gi1/0   AS-1           500a.0003.8000  Gi0/0    1s  SAC    A0001

Age of the port in the current state: 0d:00h:14m:39s

Port: Gi1/1
-----
Port state      = Up Mstr In-Bndl
Channel group = 10          Mode = Desirable-S1      Gcchange = 0
Port-channel    = Po10      GC   = 0x000A0001      Pseudo port-channel = Po10
Port index      = 0         Load = 0x00           Protocol = PAgP

Flags: S - Device is sending Slow hello.  C - Device is in Consistent state.
      A - Device is in Auto mode.          P - Device learns on physical port.
      d - PAgP is down.

Timers: H - Hello timer is running.        Q - Quit timer is running.
      S - Switching timer is running.      I - Interface timer is running.

Local information:

      Hello    Partner  PAgP    Learning  Group
Port   Flags State  Timers  Interval Count  Priority  Method  Ifindex
Gi1/1   SC   U6/S7   H       30s      1       128     Any     14

Partner's information:

      Partner          Partner          Partner          Partner Group
Port   Name            Device ID        Port    Age  Flags  Cap.
Gi1/1   AS-1           500a.0003.8000  Gi0/0    1s  SAC    A0001
```

## CCNP Switching EtherChannels Lab

```
Gi1/1      AS-1              500a.0003.8000   Gi0/1      23s SAC      A0001

Age of the port in the current state: 0d:00h:14m:34s

      Port-channels in the group:
      -----

Port-channel: Po10
-----

Age of the Port-channel   = 0d:00h:16m:42s
Logical slot/port        = 16/0              Number of ports = 2
GC                        = 0x000A0001        HotStandBy port = null
Port state                = Port-channel Ag-Inuse
Protocol                  = PAgP
Port security             = Disabled
Load share deferral      = Disabled

Ports in the Port-channel:
Index   Load   Port      EC state      No of bits
-----+-----+-----+-----+-----
   0     00    Gi1/0    Desirable-S1     0
   0     00    Gi1/1    Desirable-S1     0

Time since last port bundled: 0d:00h:14m:34s   Gi1/1
```

### Task 4: Configure Layer 2 EtherChannel LACP

#### Switch DS-1

```
interface range g0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  shutdown
  channel-protocol lacp
  channel-group 20 mode active
  no shutdown
```

#### Switch DS-2

```
interface range g0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  shutdown
  channel-protocol lacp
  channel-group 20 mode passive
  no shutdown
```

## CCNP Switching EtherChannels Lab

### Verification:

```
DS-2#show etherchannel summary
```

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby  (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
        A - formed by Auto LAG
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators:          1
```

| Group | Port-channel | Protocol | Ports                                            |
|-------|--------------|----------|--------------------------------------------------|
| 20    | Po20 (SU)    | LACP     | Gi0/0 (P)    Gi0/1 (P)    Gi0/2 (P)<br>Gi0/3 (P) |

### Task 5: Configure Layer 3 EtherChannel PAGP

#### Switch DS-2

```
interface Port-channel3
no switchport
ip address 192.168.3.1 255.255.255.0
exit
```

```
interface range g1/0-1
no switchport
channel-protocol pagp
channel-group 3 mode desirable
exit
```

#### Switch AS-2

```
interface Port-channel3
no switchport
ip address 192.168.3.2 255.255.255.0
exit
```

```
interface range g0/0-1
no switchport
```

## CCNP Switching EtherChannels Lab

```
channel-protocol pagp
channel-group 3 mode desirable
exit
```

### Verification:

DS-2#show etherchannel 3 summary

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3        S - Layer2
        U - in use        N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG
```

Number of channel-groups in use: 2

Number of aggregators: 2

| Group | Port-channel | Protocol | Ports             |
|-------|--------------|----------|-------------------|
| 3     | Po3(RU)      | PAgP     | Gi1/0(P) Gi1/1(P) |

DS-2#ping 192.168.3.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/22 ms

### Task 6: Configure Layer 3 EtherChannel LACP

#### Switch DS-1

```
interface Port-channel2
no switchport
ip address 192.168.2.1 255.255.255.0
exit
```

```
interface range g1/2-3
no switchport
channel-protocol lacp
channel-group 2 mode active
```

## CCNP Switching EtherChannels Lab

```
exit
```

### Switch AS-2

```
interface Port-channel2
no switchport
ip address 192.168.2.2 255.255.255.0
exit
```

```
interface range g0/2-3
no switchport
channel-protocol lacp
channel-group 2 mode active
exit
```

### Verification:

```
DS-1#show etherchannel 2 summary
```

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby  (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator
        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
        A - formed by Auto LAG
```

```
Number of channel-groups in use: 3
```

```
Number of aggregators:          3
```

```
Group  Port-channel  Protocol    Ports
```

```
-----+-----+-----+-----+-----+
2      Po2 (RU)      LACP       Gi1/2 (P)  Gi1/3 (P)
```

```
DS-1#ping 192.168.2.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/17/27 ms
```

### Task 7: Create static routes

Create static routes that DS-1 Po2 interface can reach DS-2 Po3 interface and vice versa.

## CCNP Switching EtherChannels Lab

### Switch DS-1

```
ip route 192.168.3.0 255.255.255.0 192.168.2.2
```

### Switch DS-2

```
ip route 192.168.2.0 255.255.255.0 192.168.3.2
```

### Verification:

```
DS-1#ping 192.168.3.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
```

```
!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 13/17/27 ms
```

```
DS-2#ping 192.168.2.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
```

```
!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 13/16/25 ms
```